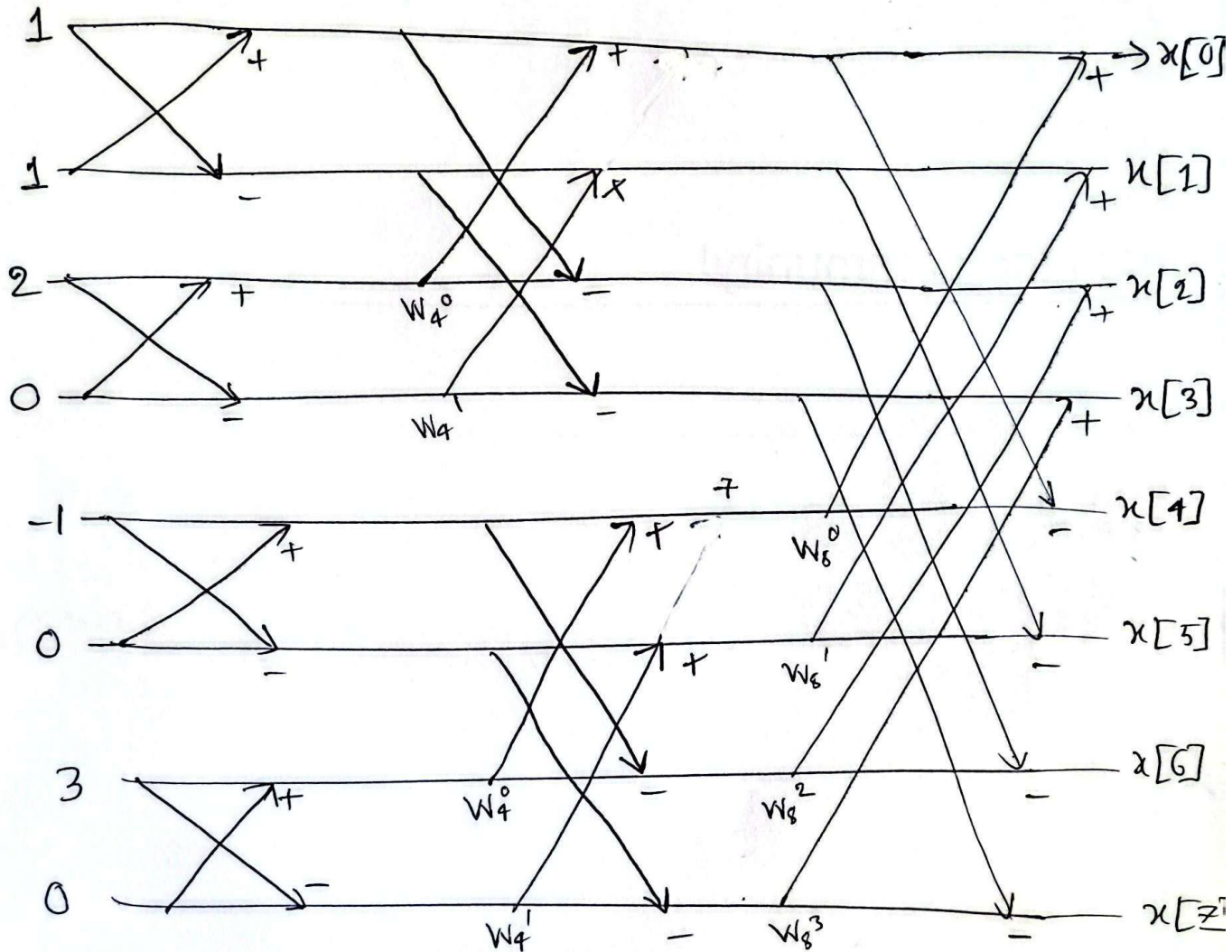


Ans-1

23+10

$$x[n] = \{ \overset{0}{\underset{\uparrow}{-1}}, \overset{1}{-1}, \overset{2}{2}, \overset{3}{3}, \overset{4}{1}, \overset{5}{0}, \overset{6}{0}, \overset{7}{0} \}$$



Ans-2

Scrambled	Binary	Reversed	true
x_0	000	000	0
x_1	001	100	4
x_2	010	010	2
x_3	011	110	6
x_4	100	001	1
x_5	101	101	5
x_6	110	011	3
x_7	111	111	7

$$x[n] = \{0, -1, 2, 2, 3, 1, 2, 0\}$$

↑

Ans-3

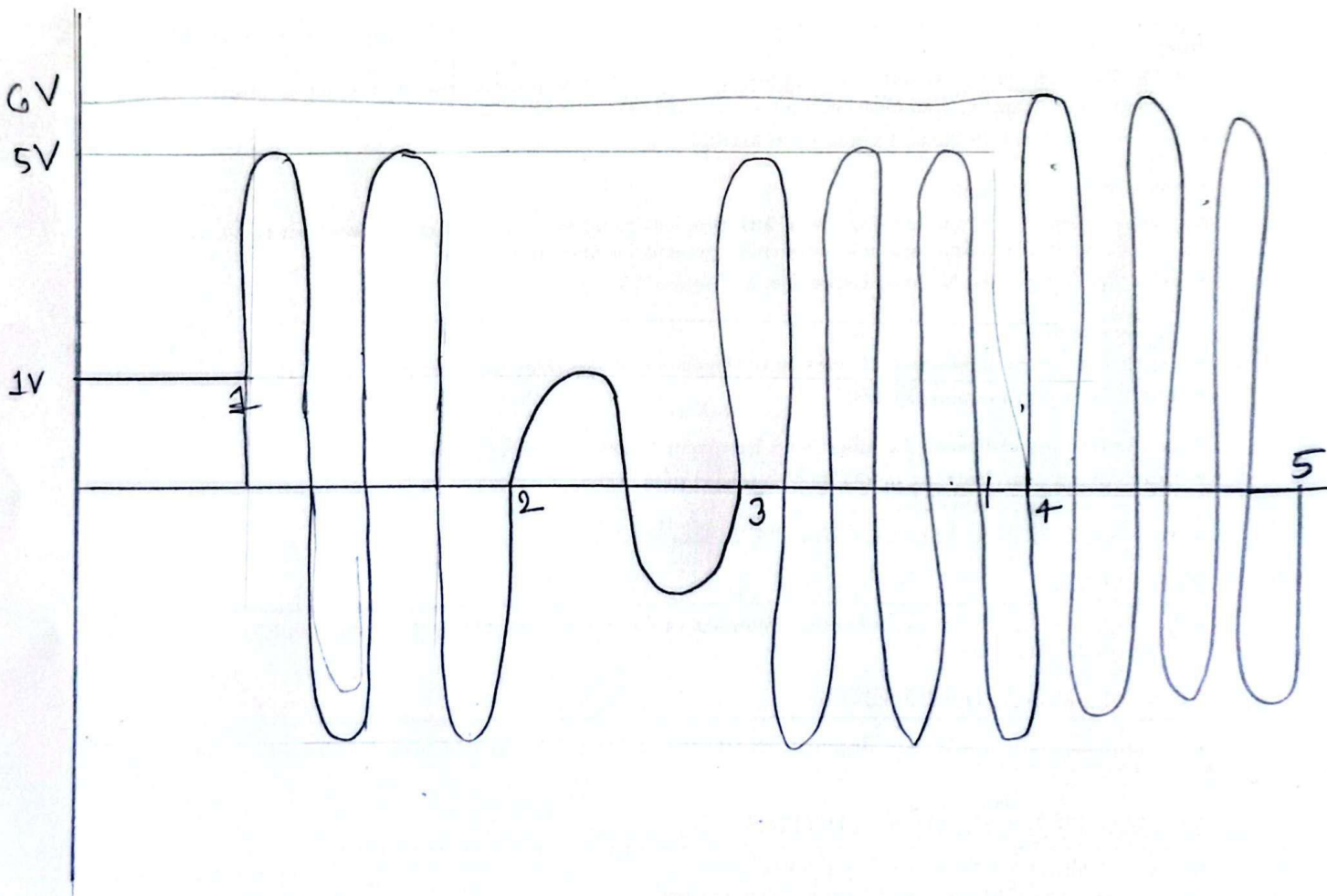
(a)

$$f_s = 2 \times f = 2 \times 5 = 10 \text{ Hz}$$

(b)

$$N = 5 \times 10 = 50$$

(c)



Frequency (kHz)	Triangular filter bank frequency weightage			
	Filter 1	Filter 2	Filter 3	Filter 4
1	0.675	0.325	0	0
2	0	0.905	0.095	0
3	0	0.451	0.548	0
4	0	0	0.999	9.615×10^{-4}
5	0	0	0.759	0.24
6	0	0	0.518	0.482
7	0	0	0.278	0.722
8	0	0	0.0375	0.9625
9	0	0	0	0.892
10	0	0	0	0.765
11	0	0	0	0.637
12	0	0	0	0.51
13	0	0	0	0.382
14	0	0	0	0.255
15	0	0	0	0.127
16	0	0	0	0

Finally, apply the weights to the time segments and derive the Mel spectrogram. Use the following formula to calculate each cell,

$$\|X_{Mel}(m, t)\| = \sum_f W^m(f) \cdot \|X(f, t)\|$$

Fill up the last table.

Mel centers		t = 0 to 1 sec	t = 1 to 2 sec	t = 2 to 3 sec
m_1	715	16.75 dB	18.09 dB	14.07 dB
m_2	1430	38.385 dB	42.2 dB	31.21 dB
m_3	2145	54.206 dB	58.364 dB	49.98 dB
m_4	2860	11.36 dB	38.602 dB	14.71 dB

Note - In the final exam you won't be guided like this. Also the question will not be this big. You will be expected to do partial calculations from any of the segments.